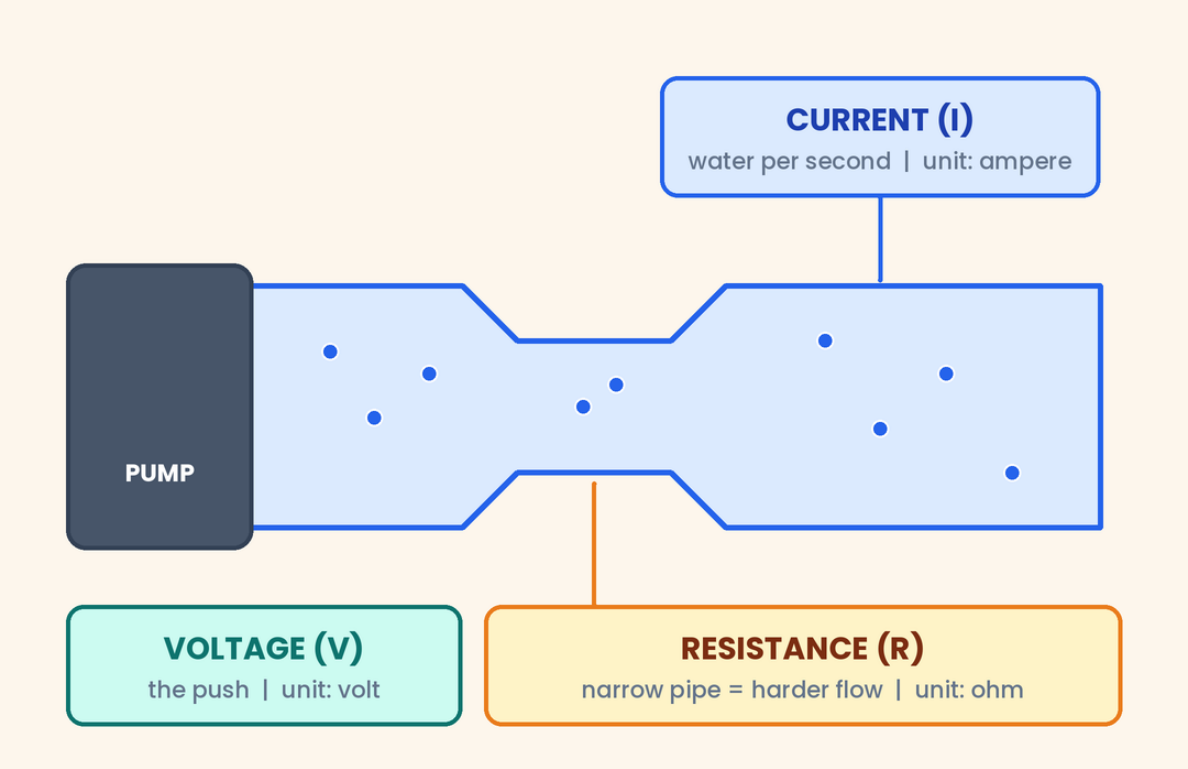
Ohm's Law

Ohm's law links three quantities of an electric circuit: voltage V (the push, unit volt), current I (the flow of charge, unit ampere) and resistance R (the opposition to flow, unit ohm, symbol Ω). A pump pushing water through a pipe is a good picture: the pump is voltage, the flow of water is current, and a narrow pipe is resistance.



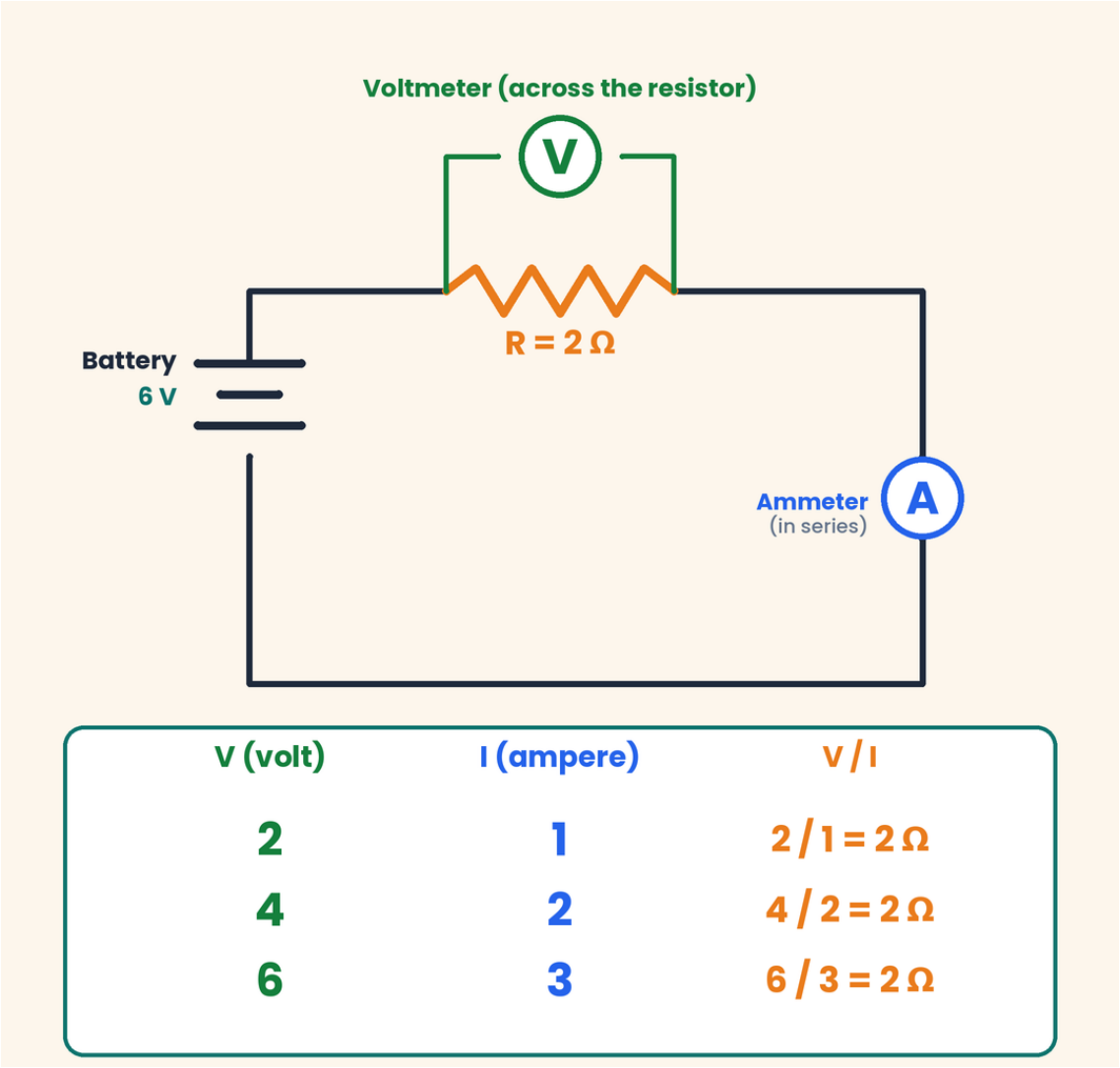
Statement (Georg Ohm, 1827)

At constant temperature, the current through a conductor is directly proportional to the potential difference across its ends: V is proportional to I, so V = I R. R is a constant for that conductor.

Formula triangle

Cover I: I = V / R. Cover V: V = I x R. Cover R: R = V / I.

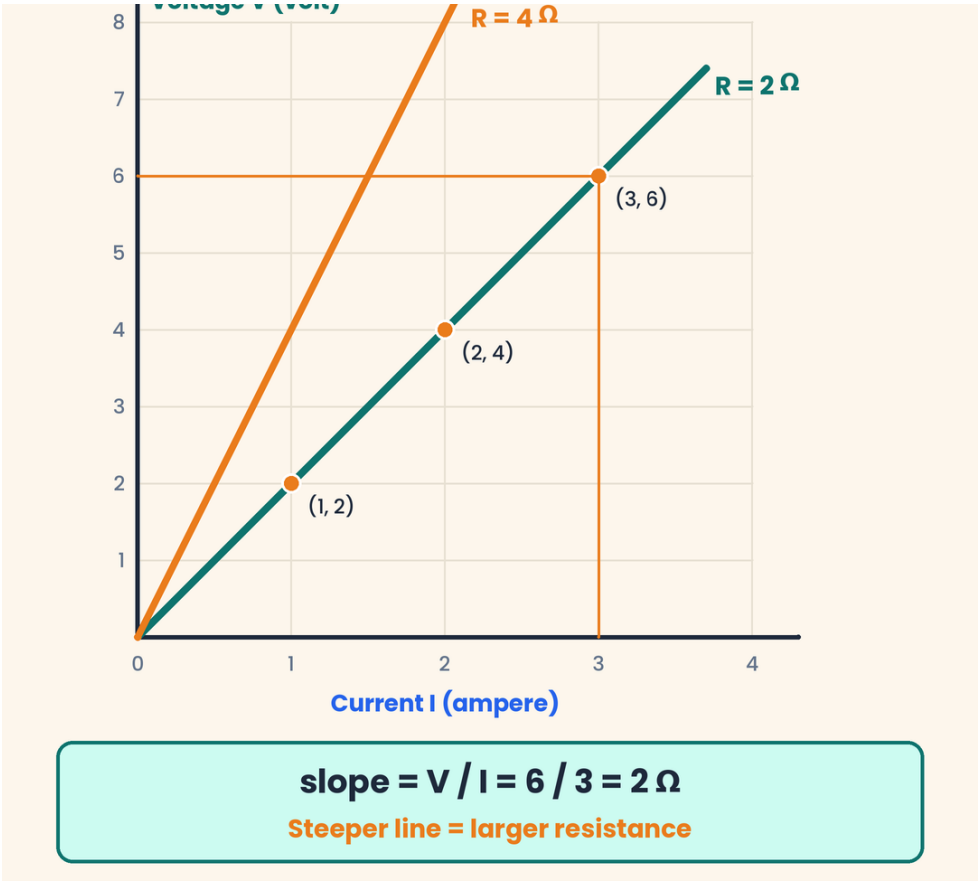
Verifying Ohm's law



V (volt)	I (ampere)	V / I
2	1	2 / 1 = 2 Ω
4	2	4 / 2 = 2 Ω
6	3	6 / 3 = 2 Ω

Connect a battery, a resistor, an ammeter in series (to measure current) and a voltmeter across the resistor. For R = 2 Ω: at 2 V the current is 1 A, at 4 V it is 2 A, at 6 V it is 3 A. V / I stays constant at 2 Ω.

The V-I graph



Plot current I on the x-axis and voltage V on the y-axis. The points lie on a straight line through the origin, so slope = V / I = R. Here slope = 6 / 3 = 2 Ω. A steeper line means a larger resistance.

Solved example

A 20 Ω resistor is connected to a 6 V battery. Find the current through it.

Given: R = 20 Ω, V = 6 V. Formula: V = I R, so I = V / R.

I = 6 / 20 = 0.3 A

Remember: Ohm's law applies only to conductors that obey it (ohmic conductors), at constant temperature.